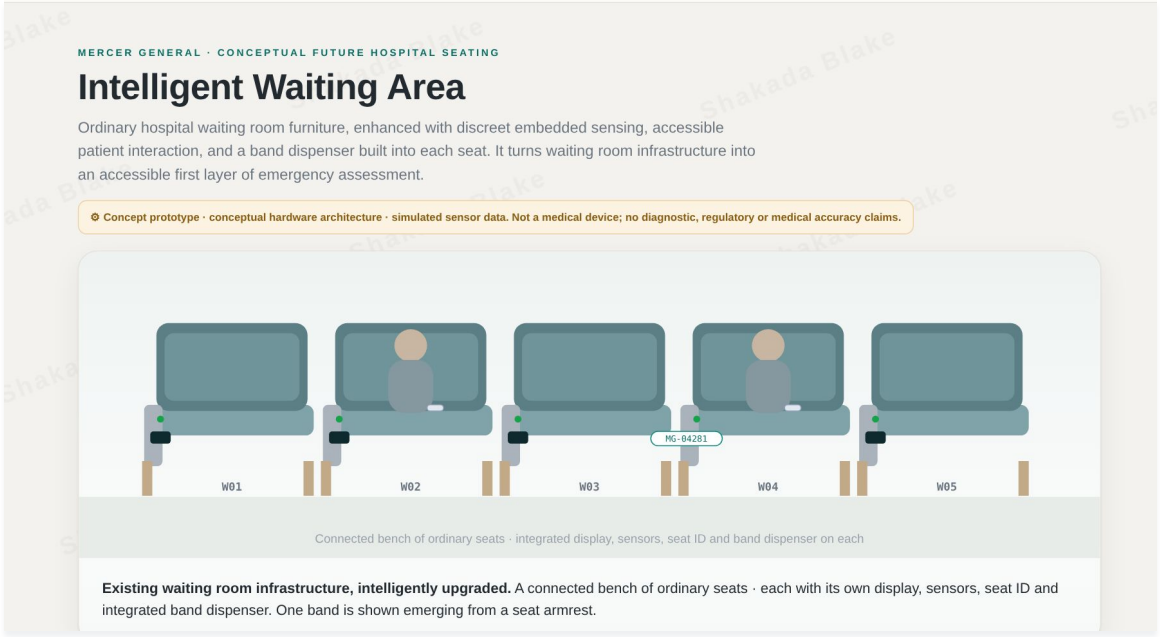
Mercer General Emergency Triage

AI supported triage from the sensing chair to the clinician, with a Gradient Boosting ESI model

Sensors inform. AI assists. Clinicians decide.

Aleksandras Group
Group presentation · CariSurg MedTech Pathways · Week 10



© 2026 Shakada Blake · CariSurg MedTech Pathways · Week 10 · Concept prototype, not a medical device

More than triage classification

The problem, beyond classification

- Patients wait largely unmonitored and can deteriorate in the queue
- Vitals are written on paper and reach the record late
- The queue is ordered by hand under surge
- So we also improve queue management, clinician cognitive load, documentation and handover quality

Why it matters in the Caribbean

- High patient to clinician ratios and staff shortages raise the value of decision support
- Real constraints: limited budgets, no bedside terminals, connectivity and power drops
- So the design is low cost, modular onto existing furniture, and safe when data is missing
- Governed under the Grenada Data Protection Act 2023, not United States rules

© 2026 Shakada Blake · CariSurg MedTech Pathways · Week 10 · Concept prototype, not a medical device

The users

- **Triage nurse: the primary user and decision maker**
- Medical assistant: intake and verified vitals
- Doctor: review, orders and disposition
- Laboratory, administrator and IT
- Patient: their own portal only
- The AI is built for the nurse, as an assistant, never the decider

Where it sits

- Triage begins in the waiting area at the sensing chairs
- It runs through triage and across the whole department
- One shared record links every role and every action
- A front door plus whole department layer, not a single screen

© 2026 Shakada Blake · CariSurg MedTech Pathways · Week 10 · Concept prototype, not a medical device

HRI · THE CHAIRS

Intelligent Waiting Area

Ordinary seats with embedded sensing, patient verification, and a QR band dispenser in each seat. Links seat, band and encounter.

THE INTELLIGENCE

Gradient Boosting ESI model

Suggests an ESI tier from triage vitals and complaint flags. Leakage removed. Decision support only.

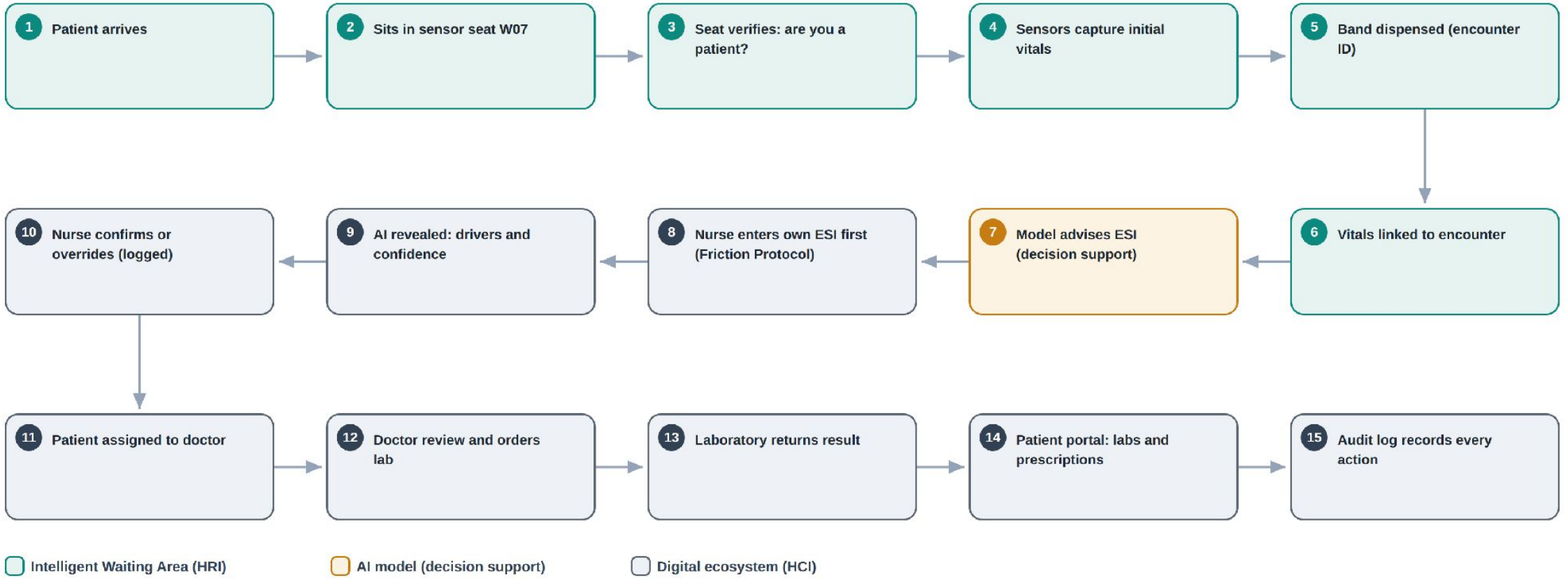
HCI · THE APP

Digital ecosystem

Multi role platform on one shared dataset, with the urgency messages, queue, and a full audit trail.

© 2026 Shakada Blake · CariSurg MedTech Pathways · Week 10 · Concept prototype, not a medical device

From the sensing chair to the audit log, on one shared record



© 2026 Shakada Blake · CariSurg MedTech Pathways · Week 10 · Concept prototype, not a medical device

Each active voice, twelve words or fewer, with at least one action verb

▲ **CRITICAL**

10 words

“See this patient immediately. Start resuscitation and alert the team.”

● **HIGH**

10 words

“Assess this patient within ten minutes. Escalate if they deteriorate.”

■ **MEDIUM**

10 words

“Examine this patient within thirty minutes. Recheck vital signs regularly.”

◆ **LOW**

9 words

“See this patient during routine rounds. Monitor for changes.”

© 2026 Shakada Blake · CariSurg MedTech Pathways · Week 10 · Concept prototype, not a medical device

Every rule the brief set, and how we met it

Twelve words or fewer

Every message is nine or ten words

Active voice, built on an action verb

See, Assess, Examine, Escalate, Monitor

Urgency escalates predictably

From "immediately" down to "routine rounds"

Rankable from the wording alone

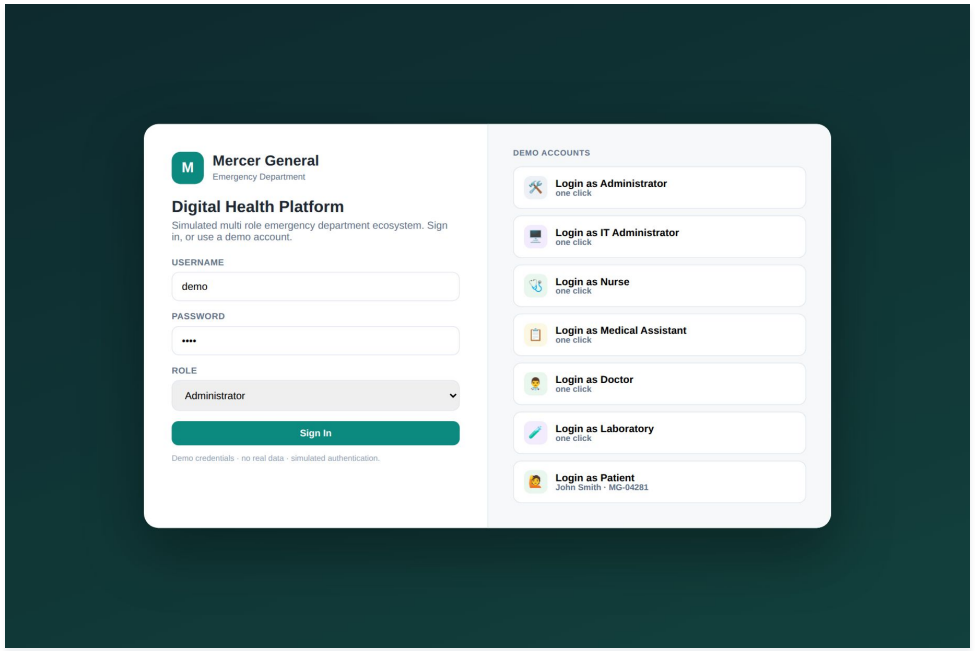
A reader can order the four with no colour

No vague verbs

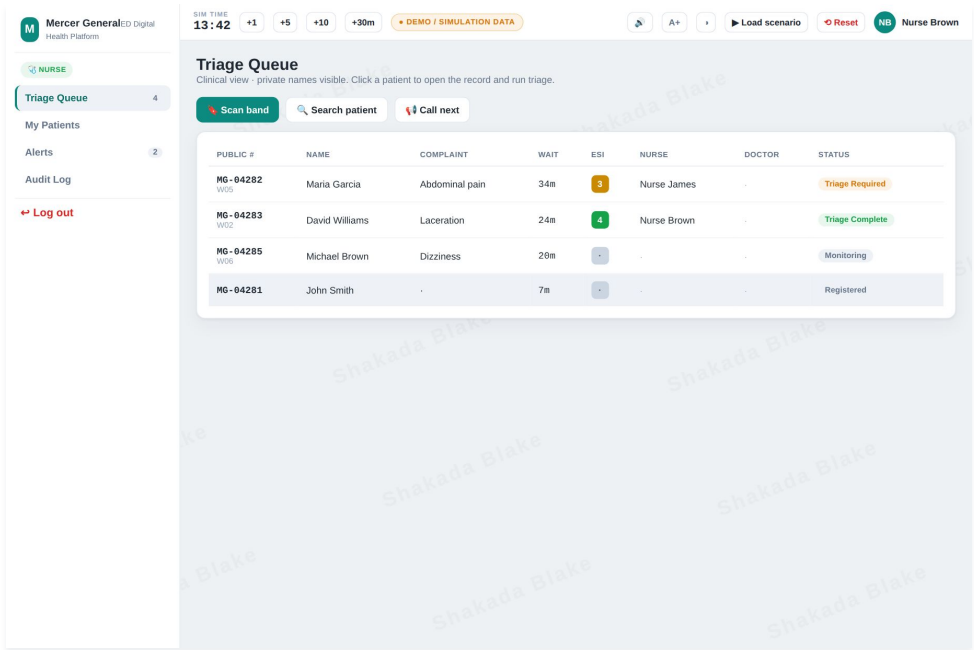
We avoided "consider" and "review" in the urgent tiers

PROTOTYPE · MY TRIAGING SYSTEM

A multi role platform on one shared record. This is my system.



Role based login across the department

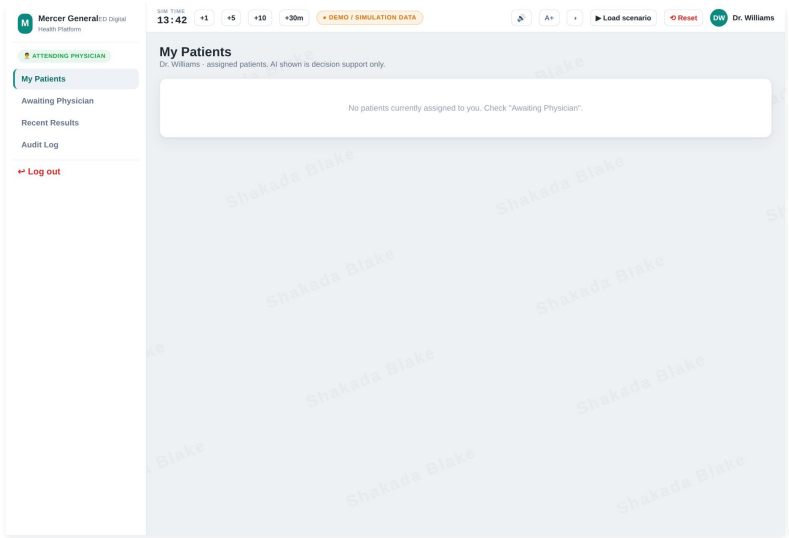


Nurse triage queue, ordered by acuity

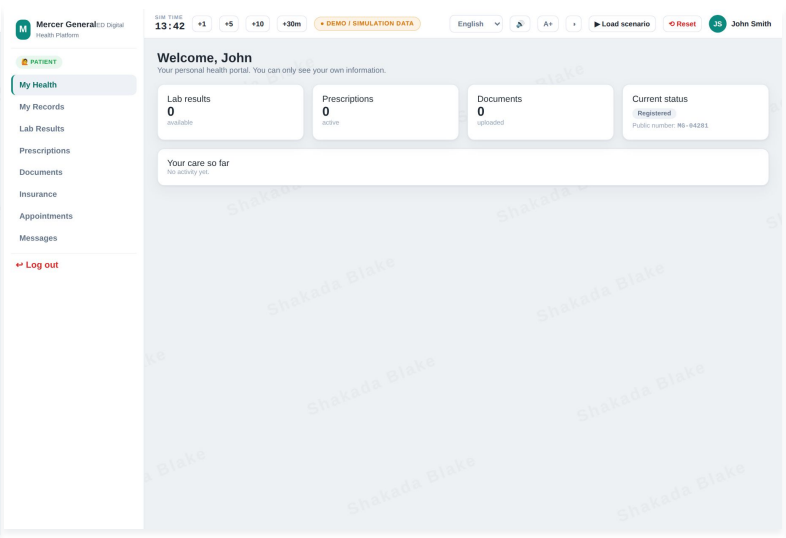
- Seven role based logins on one shared dataset: admin, IT, medical assistant, nurse, doctor, laboratory and patient
- The triage queue is ordered by acuity, with an ESI badge, public number, complaint and status for each patient
- The nurse scans the QR band to open a record and run triage, and every action is written to the audit log

PROTOTYPE · MY ECOSYSTEM DEMO

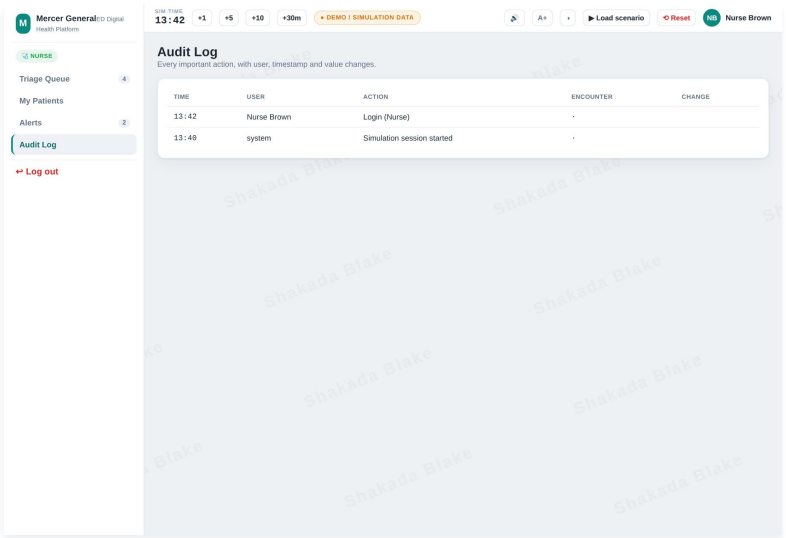
A working multi role demo: the same patient across every role, on one record



Doctor review and orders



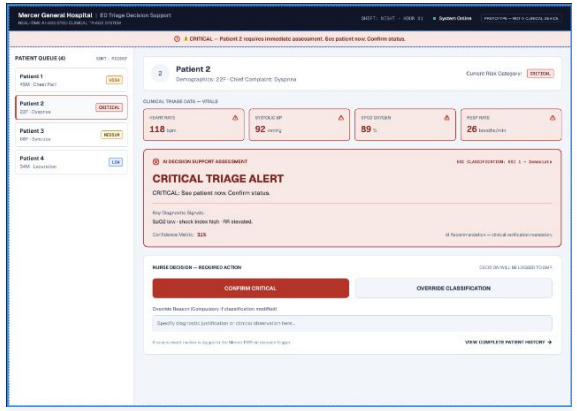
Patient portal: results and prescriptions



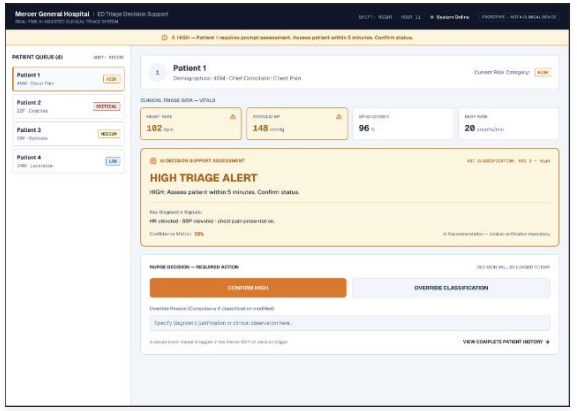
Audit log of every action

Load scenario and Reset drive a smooth live demo. A prescription a doctor writes appears in the patient portal, a lab result flows back, and every step is written to the audit log.

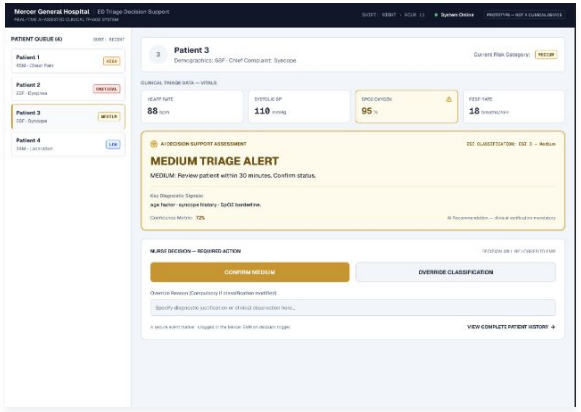
Aiden’s per tier views, incorporated into my system



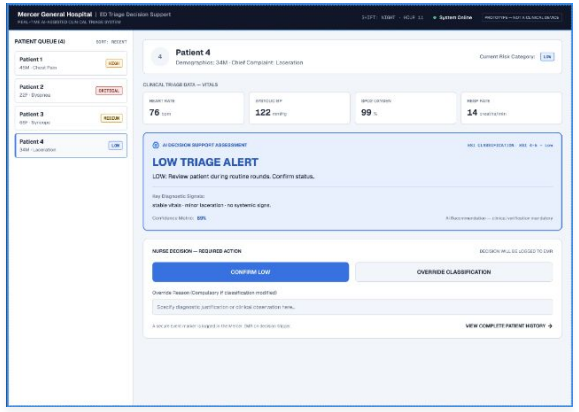
Critical



High



Medium

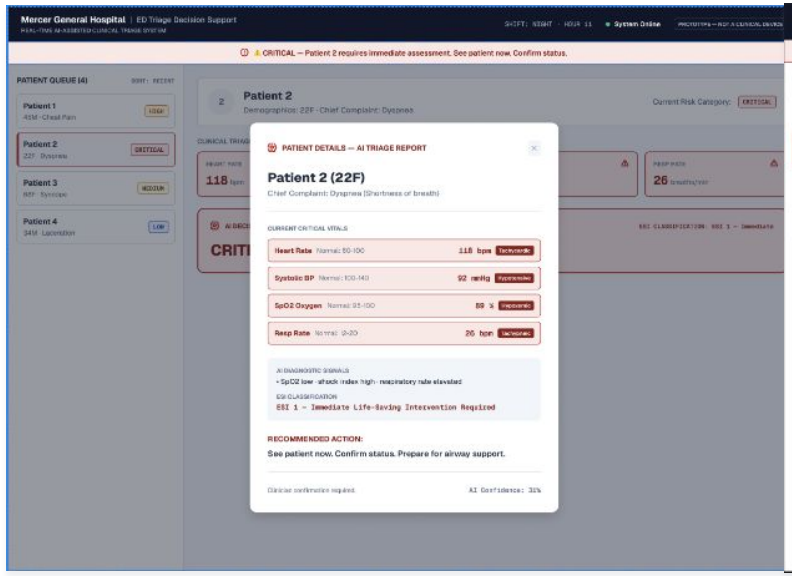


Low

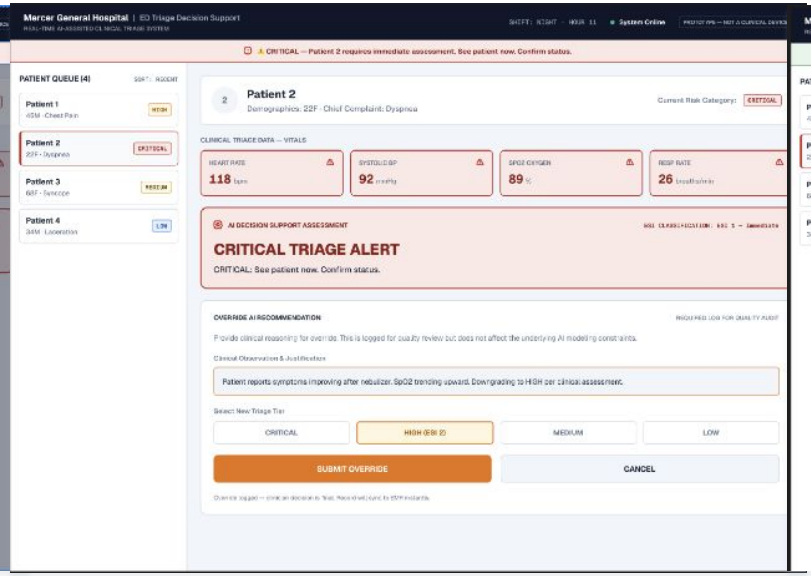
© 2026 Shakada Blake · CariSurg MedTech Pathways · Week 10 · Concept prototype, not a medical device

PROTOTYPE · VIEW DETAILS, CONFIRM OR OVERRIDE

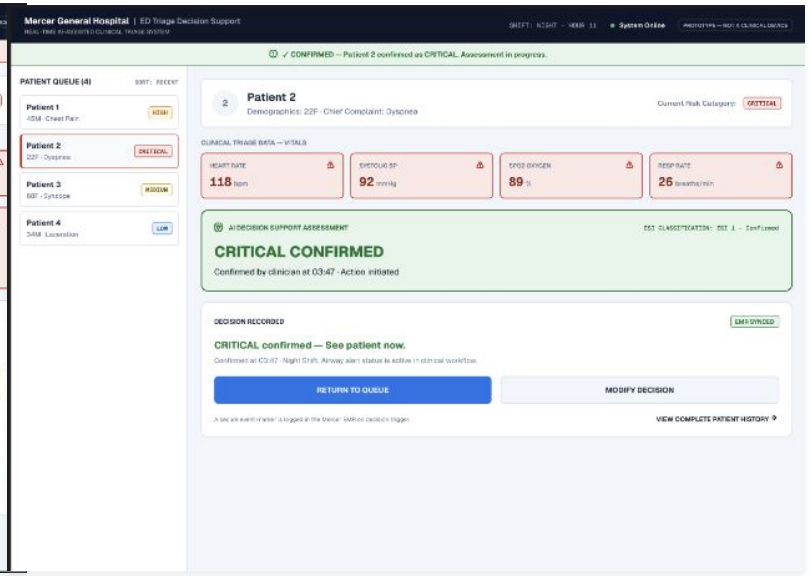
Aiden’s detail views, incorporated into my system



AI triage report with vitals and confidence



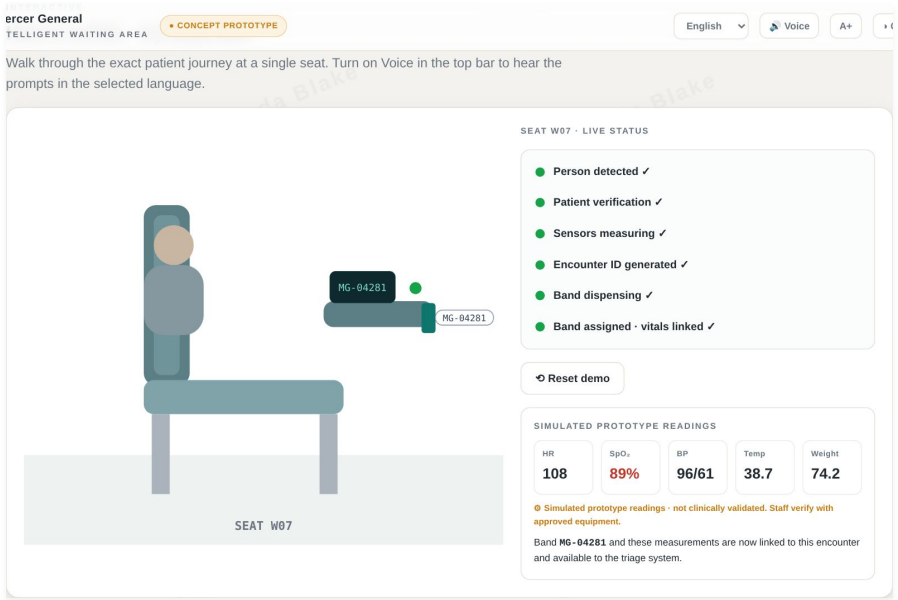
Override with a written justification

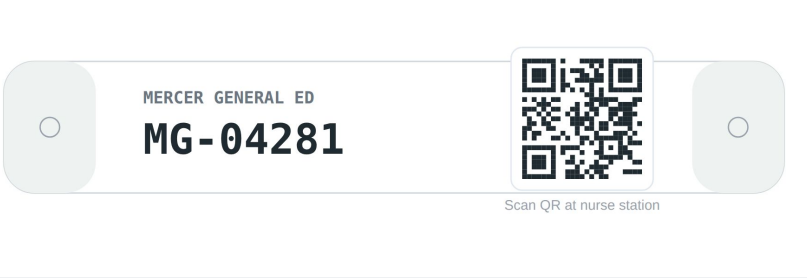


Confirmed and logged to the record

The nurse always confirms or overrides. The AI is decision support, never the final decision.

© 2026 Shakada Blake · CariSurg MedTech Pathways · Week 10 · Concept prototype, not a medical device





- Triage begins at the sensing chair: it captures initial vitals while the patient waits
- A wristband is dispensed from the seat, carrying a QR code that links seat, band, encounter and tier
- The QR code replaces the older barcode: it holds the encounter identifier and is quick to scan
- The nurse scans the QR to open the record and the urgency message

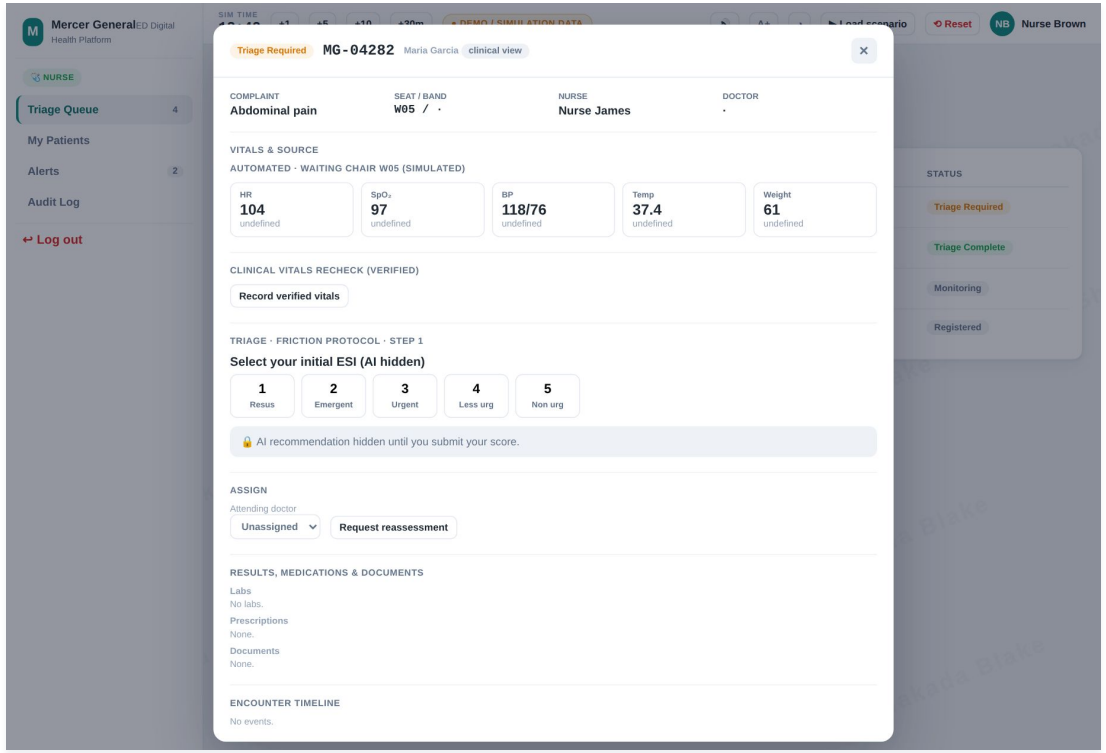
Urgency is never carried by colour alone

Tier	On screen (shape + label + colour)	Message cue	Physical cue at the chair
Critical	Filled triangle, label CRITICAL	act immediately	rapid pulsing light and spoken alert
High	Filled circle, label HIGH	within ten minutes	steady light and a single tone
Medium	Filled square, label MEDIUM	within thirty minutes	slow glow, no sound
Low	Diamond, label LOW	routine rounds	static indicator only

Every tier is signalled three ways: shape and label on screen, the wording of the message, and a distinct light and sound at the chair. A colour blind or deaf user still receives the urgency.

© 2026 Shakada Blake · CariSurg MedTech Pathways · Week 10 · Concept prototype, not a medical device

Its features and safeguards, shown in my ecosystem



The Friction Protocol in my ecosystem: the nurse scores first, the AI stays hidden.

Decision support, not decision

It suggests an ESI tier and never assigns it

Friction Protocol

The nurse records their own ESI first, the AI stays hidden

Explainability in 60 seconds

It shows the top drivers and a confidence level

Safety Hold

It abstains when key data is missing, never guesses

Human in the loop

The nurse confirms or overrides, always

Full audit trail

Every action is written to the record

Week 7 final benchmark, test set, n = 11,025, macro averaged

Model	Accuracy	Macro F1	Macro recall	Recall ESI 1
Dummy (stratified)	0.375	0.204	0.204	0.000
Logistic Regression	0.667	0.492	0.463	0.250
Decision Tree	0.556	0.216	0.245	0.000
Random Forest	0.491	0.390	0.553	0.500
Gradient Boosting	**0.668**	**0.442**	**0.435**	**0.125**

Selected model

Gradient Boosting

- Highest accuracy 0.668
- Fast inference
- Caveat: ESI 1 recall 0.125

Accuracy alone is misleading, because ESI 3 dominates the data. Gradient Boosting has the highest accuracy, but in this run it was not class balanced, so its ESI 1 recall is only 0.125. Closing that gap is my next iteration.

© 2026 Shakada Blake · CariSurg MedTech Pathways · Week 10 · Concept prototype, not a medical device

- **Gradient Boosting has the highest overall accuracy, 0.668, and fast inference**
- Accuracy is a weak guide here, because ESI 1 is only about 0.1 percent of arrivals
- Honest caveat: in this run it was not class balanced, so its ESI 1 recall is only 0.125, catching 2 of 16 critical patients, the weakest complex model
- My class balanced Random Forest caught 8 of 16, a recall of 0.500; a class balanced Gradient Boosting reached 0.4375 in an earlier run
- This is exactly why the design keeps the nurse in control and never lets the model route on its own

My next iteration: re-run Gradient Boosting with class balancing so it attends to the rare critical patients, then compare it against the class balanced Random Forest. Until the recall improves, the human in the loop is the safety net, not the model.

© 2026 Shakada Blake · CariSurg MedTech Pathways · Week 10 · Concept prototype, not a medical device

What the model uses

- Triage vitals: heart rate, blood pressure, oxygen, respiratory rate, temperature and glucose
- Chief complaint flags, for example stroke alert and abdominal pain
- 208 numeric features in total; the strongest drivers are stroke alert and blood pressure
- Configuration: histogram gradient boosting, 300 iterations, learning rate 0.08

Fairness and leakage

- Demographics excluded to reduce bias: age, gender, ethnicity, race, language, religion, marital status, employment and insurance
- Only true leakage removed: disposition and previous disposition, known only after triage
- Administrative and arrival fields also excluded
- Data: a de identified Yale triage extract, one stratified 80/20 split, seed 42

© 2026 Shakada Blake · CariSurg MedTech Pathways · Week 10 · Concept prototype, not a medical device

1. Patient sits in seat W07 and confirms they are a patient
2. Initial vitals are captured and a QR band, MG-04281, is dispensed
3. The nurse calls the number to the public display, then scans the QR
4. The record opens; the nurse records their own ESI first
5. The model reveals a suggested tier with its drivers and confidence
6. The nurse confirms or overrides with a written reason
7. The patient is assigned to a doctor and the status updates everywhere
8. The audit log records the full timeline

© 2026 Shakada Blake · CariSurg MedTech Pathways · Week 10 · Concept prototype, not a medical device

ETHICS AND SAFETY

Bias, accountability, and the three biggest risks

Ethical foundations

- Automation bias, countered by the Friction Protocol
- Accountability: the clinician decides, the audit log records it
- Consent: the chair asks before it acts; the portal controls sharing
- Data governance: minimisation, fairness, Grenada DPA 2023

Three safety risks and residual risk

1. Under triage of a critical patient

Nurse decides, acuity ordering, deterioration alerts. Residual: the model can still under flag.

2. Over reliance under pressure

Friction Protocol and logged overrides. Residual: a rushed nurse may still defer.

3. Missing data or power and connectivity loss

Safety Hold abstains, manual intake always works. Residual: sensors are not medical grade.

© 2026 Shakada Blake · CariSurg MedTech Pathways · Week 10 · Concept prototype, not a medical device

Method used

- **Moderated think aloud usability test with a blind ranking task**
- Colour was removed, so testers judged urgency from the wording alone
- Sessions were captured as verbatim notes, not summaries
- Three peer testers, each seen individually

What each tester was asked to do

- Rank the four messages from most to least urgent
- Explain, in their own words, what each message told them to do
- Point out any wording they found confusing or ambiguous
- Suggest any change they would make

© 2026 Shakada Blake · CariSurg MedTech Pathways · Week 10 · Concept prototype, not a medical device

1. Colour blindness

Mechanism: Every tier carries a distinct shape and a text label as well as colour, and the messages rank from wording alone.

Honest response: shapes were chosen for distinctness but not yet verified with a colour blindness simulator or affected users.

2. Cognitive load

Mechanism: Messages are twelve words or fewer with one clear action, a consistent structure, and an alert banner for the single most urgent patient.

Honest response: not yet tested under a simulated surge; a long queue could still overwhelm a busy nurse.

3. Ambient noise at the chair

Mechanism: The chair spoken prompt is mirrored by on screen text and a visual cue, so audio is never the only channel.

Honest response: audio legibility in a noisy department is not yet measured; volume and direction need testing.

© 2026 Shakada Blake · CariSurg MedTech Pathways · Week 10 · Concept prototype, not a medical device

The report (about three pages, British English)

- **a. Design decisions**
- Why these messages, and why a screen based prototype paired with the sensing chair
- **b. Testing methodology and results**
- The think aloud ranking task, three testers, verbatim
- **c. Intended next iteration**
- The specific changes planned from the feedback

Next iteration (specific)

- Verify the shape and label icons with a colour blindness simulator
- Re run the blind ranking task after any wording change
- Add an audible cue for Critical at the chair and test against recorded noise
- Trial the Medium wording for the thirty minute expectation
- Run the test with three real clinicians and capture verbatim feedback
- Re-run Gradient Boosting with class balancing to lift ESI 1 recall, and compare against Random Forest

© 2026 Shakada Blake · CariSurg MedTech Pathways · Week 10 · Concept prototype, not a medical device

Current limitations and accessibility gaps

- Proof of concept: ESI 1 recall is only 0.125 in this run, so the model misses most critical patients without class balancing; trained on the Yale extract, not local data
- Prototypes are front end, with scripted AI and conceptual chair sensing
- No real record integration or production security yet
- Cannot be used by patients who are unconscious, confused or too unwell to sit and self verify
- Stretcher patients bypass the chair; languages beyond the five supported; low digital literacy

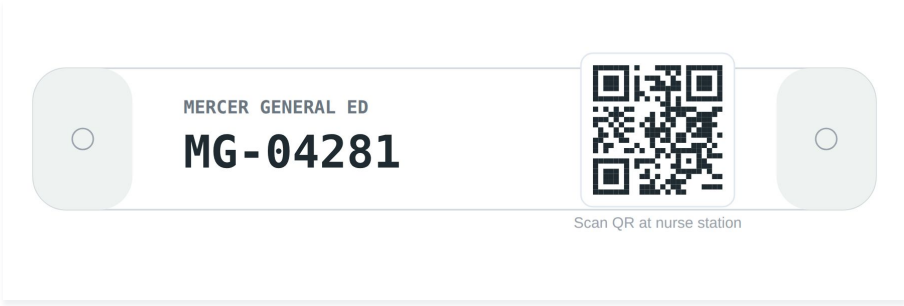
To reach a supervised pilot

- Revalidate on de identified local data, with a bias audit on the local population
- A DPIA and sign off from a Data Protection Officer and legal counsel
- Security, authentication and real record integration
- Validate any sensor hardware
- A shadow mode evaluation against nurse decisions, then ethics approval, governance and training

© 2026 Shakada Blake · CariSurg MedTech Pathways · Week 10 · Concept prototype, not a medical device

CONCLUSION

- Four urgency messages a reader can rank from the wording alone, no colour needed, shown in context
- A Gradient Boosting ESI model, selected for its overall accuracy and used only as decision support, with class balancing to lift critical recall as the next step
- Rooted in the triage process from the sensing chair and its QR band
- Tested with peers, honest about accessibility gaps, with a clear next iteration
- Built for the real constraints of a Caribbean emergency department



Sensors inform. AI assists. Clinicians decide.